BRITISH STANDARD

BS EN 50121-1:2000

Railway applications — Electromagnetic compatibility —

Part 1: General

The European Standard EN 50121-1:2000 has the status of a British Standard

ICS 29.280; 33.100.10; 45.020

NO COPYING WITHOUT BSI PERMISSION EXCEPT AS PERMITTED BY COPYRIGHT LAW

National foreword

This British Standard is the official English language version of EN 50121-1:2000. It supersedes DD ENV 50121-1:1996 which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee GEL/9, Railway electrotechnical applications, which has the responsibility to:

— aid enquirers to understand the text;

— present to the responsible European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;

— monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this committee can be obtained on request to its secretary.

Cross-references

The British Standards which implement international or European publications referred to in this document may be found in the BSI Standards Catalogue under the section entitled "International Standards Correspondence Index", or by using the "Find" facility of the BSI Standards Electronic Catalogue.

A British Standard does not purport to include all the necessary provisions of a contract. Users of British Standards are responsible for their correct application.

Compliance with a British Standard does not of itself confer immunity from legal obligations.

Summary of pages

This document comprises a front cover, an inside front cover, the EN title page, pages 2 to 15 and a back cover.

The BSI copyright notice displayed in this document indicates when the document was last issued.

This British Standard, having been prepared under the direction of the Electrotechnical Sector Committee, was published under the authority of the Standards Committee and comes into effect on 15 December 2000

© BSI 12-2000

ISBN 0 580 36752 5

Amendments issued since publication

Amd. No.	Date	Comments

EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN 50121-1

September 2000

ICS 29.020; 29.280; 45.020

English version

Railway applications - Electromagnetic compatibility - Part 1: General

Applications ferroviaires - Compatibilité électromagnétique Partie 1: Généralités

Bahnanwendungen - Elektromagnetische Verträglichkeit Teil 1: Allgemeines

This European Standard was approved by CENELEC on 2000-04-01. CENELEC members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration.

Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Central Secretariat or to any CENELEC member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CENELEC member into its own language and notified to the Central Secretariat has the same status as the official versions.

CENELEC members are the national electrotechnical committees of Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and United Kingdom.

CENELEC

European Committee for Electrotechnical Standardization
Comité Européen de Normalisation Electrotechnique
Europäisches Komitee für Elektrotechnische Normung

Central Secretariat: rue de Stassart 35, B - 1050 Brussels

© 2000 CENELEC - All rights of exploitation in any form and by any means reserved worldwide for CENELEC members.

Ref. No. EN 50121-1:2000 E

Foreword

This European Standard was prepared by the Technical Committee CENELEC TC 9X, Electrical and electronic applications for railways.

The text of the draft was submitted to the formal vote and was approved by CENELEC as EN 50121-1 on 2000-04-01.

The following dates were fixed:

- latest date by which the EN has to be implemented at national level by publication of an identical national standard or by endorsement (dop) 2001-04-01

- latest date by which the national standards conflicting with the EN have to be withdrawn (dow) 2003-04-01

Annexes designated "normative" are part of the body of the standard.

Annexes designated "informative" are given only for information.

In this standard, annex A is informative and annex B is normative.

© BSI 12-2000

Contents

Introduction 4
1 Scope 5
2 Normative references 6
3 Definitions 6
4 Performance criteria 6
5 Management of EMC 7

Annex A (informative) The railway system 8
Annex B (normative) Management of EMC at the interface between railway infrastructure and trains 13

© BSI 12-2000

Introduction

The railway EMC set of product-specific European Standards is intended, in the main, to permit compliance with the EMC Directive. It consists of five parts described at the end of this introduction.

The set of standards specifies the limits for the electromagnetic (EM) emission of the railway as a whole to the outside world and of the EM emission and immunity for equipment operating within the railway but which must be compatible with the emission limits set for the railway as a whole. The frequency covered by the standards is in the range of d.c. to 400 GHz. At present testing is not defined for frequencies exceeding 1 GHz. The limits for EMC phenomena are set so that the railway as a whole satisfies the EU Directive 89/336 on electromagnetic compatibility, and so that EMC is achieved between the various parts of the railway. Throughout the set of the standards the immunity levels are chosen to ensure a reasonable level of EMC with other apparatus within the local railway Environment and with emissions which enter the railway from the outside world. Limits are also placed on EM emission by railways into the outside world.

The immunity and emission levels do not of themselves guarantee that integration of the apparatus will necessarily be satisfactory. The standard cannot cover all the possible configurations of apparatus, but the test levels are sufficient to achieve satisfactory EMC in the majority of cases. In exceptional circumstances, for instance near a "special location" (as defined in the EMC Directive) which has unusually high levels of EM interference, the system may require additional measures to be taken to ensure proper operation. The resolution of this is a matter for discussion between the equipment supplier and the project manager, infrastructure controller or equivalent.

The railway apparatus is assembled into large systems and installations, such as trains and signalling control centres. Details are given in annex A. It is not, therefore, possible to establish immunity tests and limits for these large assemblies. The immunity levels for the apparatus will normally ensure reliable operation, but it is necessary to prepare an EMC management plan, to deal with specific circumstances. For example, the passage of the railway line close to a higher power radio transmitter which produces abnormally high field strengths. Special conditions may have to be applied for railway equipment which has to work near such a transmitter and these will be accepted as National Conditions for the specification.

The series of standards EN 50121, Railway applications - Electromagnetic compatibility, contains the following parts:

Part 1: General

This part gives a description of the electromagnetic behaviour of a railway; it specifies the performance criteria for the whole set. A management process to achieve EMC at the interface between the railway infrastructure and trains is provided.

Part 2: Emission of the whole railway system to the outside world

This part sets the emission limits from the railway to the outside world at radio frequencies. It defines the applied test methods and gives information on typical field strength values at traction and radio frequency (cartography).

Part 3-1: Rolling stock - Train and complete vehicle

This part specifies the emission and immunity requirements for all types of rolling stock. It covers traction stock and trainsets, as well as independent hauled stock.

The scope of this part of the standard ends at the interface of the stock with its respective energy inputs and outputs.

Part 3-2: Rolling stock - Apparatus

This part applies to emission and immunity aspects of EMC for electrical and electronic apparatus intended for use on railway rolling stock.

© BSI 12-2000

Part 4: Emission and immunity of the signalling and telecommunications apparatus

This part specifies limits for electromagnetic emission and immunity for signalling and telecommunications apparatus.

Part 5: Emission and immunity of fixed power supply installations and apparatus

This part applies to emission and immunity aspects of EMC for electrical and electronic apparatus and components intended for use in railway fixed installations associated with power supply.

EN 50121-1 and EN 50121-2 are product family standards which take precedence over generic standards.

EN 50121-3-1, EN 50121-3-2, EN 50121-4, and EN 50121-5 are Product Standards.

1 Scope

1.1 This Part 1 of the European Standards series EN 50121 outlines the structure and the content of the whole set.

Annex A describes the characteristics of the railway system which affect electromagnetic compatibility (EMC) behaviour. Annex B specifies a management process for achieving EMC at the interface between the railway infrastructure, as defined in the EU Directive 91/440/EEC, and trains.

The objective of the whole set of standards is to specify the EM emission and immunity requirements for railway products, and for the railway as an installation.

Phenomena excluded from the set are Nuclear EM pulse, abnormal operating conditions and the induction effects of direct lightning strike.

Emission limits at the railway boundary do not apply to intentional transmitters within the railway boundaries.

The biological effects of non-ionizing radiation as well as apparatus for medical assistance, such as pacemakers, are not considered here.

1.2 This European Standard is supplemented by the following specific standards.

EN 50121-2	Railway applications - Electromagnetic compatibility - Part 2: Emission of the whole railway system to the outside world.
EN 50121-3-1	Railway applications - Electromagnetic compatibility - Part 3-1: Rolling stock - Train and complete vehicle.
EN 50121-3-2	Railway applications - Electromagnetic compatibility - Part 3-2: Rolling stock – Apparatus.
EN 50121-4	Railway applications - Electromagnetic compatibility - Part 4: Emission and immunity of the signalling and telecommunications apparatus.
EN 50121-5	Railway applications - Electromagnetic compatibility - Part 5: Emission and immunity of fixed power supply installations and apparatus.

© BSI 12-2000

2 Normative references

This European Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate place in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

EN 61000-6-2	Electromagnetic compatibility (EMC) - Part 6-2: Generic standards - Immunity for industrial environments (IEC 61000-6-2).
IEC 60050(161)	International Electrotechnical Vocabulary (IEV) - Chapter 161: Electromagnetic compatibility.

3 Definitions

For the purpose of this European Standard definitions related to EMC and to relevant phenomena may be found in IEC 60050-161.

Other parts of this European Standard may contain specific definitions.

4 Performance criteria

NOTE This clause is based on EN 61000-6-2.

The variety and the diversity of the apparatus within the scope of this set of standards makes it difficult to define precise criteria for the evaluation of the immunity test results.

If, as a result of the application of the tests defined in this set of standards, the apparatus becomes dangerous or unsafe, the apparatus shall be deemed to have failed the test.

A functional description and a definition of performance criteria, during or as a consequence of the EMC testing, shall be provided by the manufacturer and noted in the test report, based on the following criteria.

Performance criterion A: The apparatus shall continue to operate as intended during and after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation, and from what the user may reasonably expect from the apparatus if used as intended.

Performance criterion B: The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is however allowed. No change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation, and from what the user may reasonably expect from the apparatus if used as intended.

Performance criterion C: Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls.

© BSI 12-2000

5 Management of EMC

The railway is a complex installation with moving sources of electromagnetic energy and the application of the EMC standards in the EN 50121 series is not a guarantee of satisfactory performance. There may be cases where apparatus has to be positioned in restricted spaces or added to an existing assembly, with the possible creation of environments of unusual severity. All cases shall be considered with respect to a formal plan for the management of EMC. This plan should be established at as early a stage of the project as is possible and annex B gives detail of a management plan.

NOTE Annex B is normative with a limited scope; however, operation outside the scope should consider whether the use of the management plan could be beneficial.

© BSI 12-2000

Annex A (informative)

The railway system

A.1 Introduction

For operating purposes railways use electrical systems that require very high outputs (up to several MVA) and power electronic systems that are characterized by their non-linearity (producing harmonics).

In an electric railway the locomotives which move with the trains have to be supplied via sliding contacts from a supply line, called the catenary or overhead, or a trackside conductor rail, which is installed along the track. The current generally returns to the substation via the rails, a separate return conductor or via the earth. The railway is an integrated system in which electricity has many uses in addition to train propulsion including:

- heating, air conditioning, catering and lighting of passengers coaches with converters on the vehicles. This power is fed along the train by separate conductors;
- signalling and telecommunication systems along the track and between control centres, concerned with the movement of trains;
- computer installations in control centres, linked via trackside routes;
- passenger information systems on vehicles, stations and depots;
- traction within diesel-electric locomotives and multiple units;
- battery traction vehicles.

Hence problems of EMC arise not only within the locomotive and the power supply but also in these associated systems. Non-electrified traction such as diesel electric traction may also be a source of EM noise.

The normal and disturbed working of these systems may be a source of electromagnetic noise which can affect all other systems.

A.2 General coupling mechanisms

The coupling between systems is by the well known physical phenomena, and limits are expressed in terms of these phenomena.

Five modes of coupling are distinguished:

- electrostatic coupling, in which a charged body is discharged to a victim circuit;
- capacitive coupling, in which the varying voltage in one circuit produces voltage changes in a victim circuit via mutual capacitance;
- Inductive coupling, in which a varying magnetic field produced by one current in one circuit, links with a victim circuit, inducing a voltage via mutual inductance;
- conductive coupling, in which the source and victim circuits share a common conduction path;
- electric (E) and magnetic (H) radiation, in which the circuit structures act as antenna transmitting and receiving energy.

A.3 Principal electromagnetic phenomena for immunity

A.3.1 Conducted low frequency phenomena

Slow variations of the supply voltage, including dips, surges, fluctuations, unbalance. Harmonics and intermodulation products. Data transfer carried on the power supply. Power frequency variations, induced low frequency voltages, d.c. in a.c. networks.

© BSI 12-2000

A.3.2 Low frequency field phenomena

Magnetic fields, both steady and transient. Electric fields.

A.3.3 Conducted high frequency phenomena

Unidirectional and oscillatory transients, as single events or repetitive bursts. Induced currents.

A.3.4 Radiated high frequency phenomena

Magnetic fields. Electric fields. Radio frequency radiated waves.

A.4 Principal electromagnetic phenomena for emission

In principle the same phenomena exist as are listed for immunity, but at this moment limits have only been applied to the following:

- magnetic fields produced by power frequency and harmonic frequency currents, up to 9 kHz;
- voltage fluctuations produced by power frequency and harmonic currents;
- radio frequency fields produced by trains.

A.5 Description of the different electric traction systems

Direct current and alternating current sources are used.

d.c. systems include:

High voltage	:	3 000 V
Medium voltage	:	1 500 V
Low voltage	:	from 600 V to 1 400 V, including more particularly urban transit systems.

a.c. systems include:

Industrial frequency	:	50 Hz at 25 kV or Autotransformer 50/25 kV
Low frequency	:	16,7 Hz at 15 kV.

Isolated three phase lines exist with two overhead conductors.

A.6 Components of electric traction systems

Traction power is generally supplied from the high voltage national or railway grid systems at voltages up to 400 kV. Connection points, known as sub-stations, perform the following functions:

- protection (circuit breakers) for both public and railway interests;
- adaptation of voltage level by transformer;
- possible rectification to provide d.c. supply or frequency conversion to give low frequency supply.

The power obtained by this means is transmitted to the traction vehicle via a system of flexible-suspension contact lines (known as the overhead catenary) with which a locomotive-mounted articulated device (known as the pantograph) is brought into contact. On low voltage lines, a trackside conductor rail may be provided from which power is collected by a sliding contact (known as the collector shoe).

On the traction vehicle the power is regulated and supplied to electric motors to control the movement of the train. Auxiliary power is also regulated and although of lower power than that supplied to the electric traction motors, can still be a significant source of electromagnetic noise.

© BSI 12-2000

On a.c. lines, circuit components may be added to the traction supply lines (auto-transformers or booster transformers) to reduce the magnetic field and hence the induced voltage in telecommunication circuits.

A.7 Internal sources of electromagnetic noise

There are several rail-specific components which produce electromagnetic noise. These include:

A.7.1 Static elements

The overhead line of the railway and the high voltage line feeding the substation can be the source of high or low frequency noise.

Among the phenomena which are involved in RF emission are:

- the corona effect, where ionization of neutral molecules in the electric field close to the conductors produces RF noise. This can exist along the whole alignment;
- brush discharges in zones of high voltage gradient on the surface of insulators;
- discharge type micro-arcs at bad contacts between energized metallic parts. These effects are local and attenuate rapidly with distance;
- partial flashovers across dry bands of polluted insulator surfaces.

Railway overhead systems differ from most high voltage overhead lines by being closer to the ground, having more insulators and having less natural cleaning of the insulators.

Low frequency noise can be significant within a wide zone, up to 3 km (or more if the ground resistivity is high). It is produced transiently at substations when high voltage switching takes place, is distributed along the overhead when it is energized, is enhanced when non-linear traction loads such as rectifiers are supplied, and is stimulated locally when flashover takes place. If a d.c. traction system is used, low frequency harmonics are produced by the rectifier substation.

A.7.2 Mobile elements

Motive power units (electric locomotives or multiple unit coaches) are a source of electromagnetic noise during routine working, primarily controlled by the following equipment:

- power control systems using controlled semiconductors such as thyristors, GTO's and IGBT's. These produce energy, which give either direct radiation from the vehicle components or indirect radiation via the power supply lines. An overhead line can act as an antenna;
- auxiliary apparatus on traction vehicles may have relatively high power rating and must be considered as a source of noise;
- the sliding contact between the line and the pantograph (or shoe and rail). This collection is via a series of short arcs which act as radio sources;
- special case arcing and transients which are produced when the pantograph is raised or lowered, or the vehicle circuit breaker is closed or opened.

Diesel-electric locomotives should be included in study since they can contain semiconductor power control which can generate noise. Such locomotives also contain auxiliary systems which may be sources.

A.7.3 Auxiliary power converters

Coaching stock air conditioning, catering and similar systems may be supplied via a semiconductor static converter and this may be sources of noise. These converters may be on several coaches in a train and the summation of their noise must be considered.

© BSI 12-2000

A.7.4 Train line

The locomotive supplies power, generally at voltages less than or equal to 1 500 V, sometimes at 3 000 V, at powers up to 800 kW, to the electric systems of the train for lighting, heating, air-conditioning, batteries, and converters through a conductor (termed "train line"). This current, which can be 800 A, is a source of noise to adjacent equipment.

This auxiliary current may return to the locomotive via the rails and hence have an influence on apparatus on the track. Train lengths of several hundred metres are not unusual.

A.7.5 Traction return current with respect to track circuits

An electrical supply (continuous, alternating or pulse) is connected across the running rails, in what is known as a track circuit. When a train travels on the track, its axles short-circuit a detector of this electrical supply and the presence of the train is detected. Electrical noise may energize the detector although the train is present, giving a false indication of clear track. Track circuits take many forms with some having frequency and time coding to reduce the risk of false energization.

Since the power supply may contain voltage components at track circuit frequencies, the input impedance of the train may have to be greater than a specified value. This prevents the passage of currents at track circuit frequencies in the running rails. The traction and auxiliary equipment on the vehicle and the substations shall not generate currents at track circuit frequencies which exceed specified values. Limits are applied for particular cases. These effects are entirely internal to the railway and many different cases can exist.

A.7.6 Trackside equipment

Electricity is used in trackside cabinets to drive switch motors, heating and train pre-heating as well as other apparatus. Although of relatively low power, these elements are close to the line and may affect other railway apparatus.

A.8 Summary of main characteristics of railways

The essential differences between electric railways and other large electric networks are:

- a very wide variety of power supply configurations;
- a very wide variety of power use and control systems and sub-systems;
- the use of sliding contacts to convey high powers to the moving trains;
- the high speed of some trains;
- the presence of several moving sources within the same zone of influence;
- a fluctuating and imprecise system of current flow to and from the train, including the passage of current via the ground;
- high single phase loads which may cause imbalance in the three phase system;
- the possibility of simultaneous generation of disturbance from several sources;
- generation of EM noise over a wide frequency spectrum;
- the interaction of supply and vehicles to enhance or diminish the effect at any given frequency.

© BSI 12-2000

A.9 External sources of disturbance

The railway is distributed through the public domain and is exposed to various sources of EM noise at various places.

These include:

- neighbouring railway systems;
- trackside radio stations, sometimes operating at high powers;
- portable radio transmitters including portable telephones;
- adjacent overhead power lines from which power frequency induction may be experienced;
- radar sets at airports, on aircraft, in military use;
- industrial plants which disturb the electricity supply network.

© BSI 12-2000

Annex B (normative)

Management of EMC at the interface between railway infrastructure and trains

B.1 Introduction

The EU Directive 91/440/EEC on the development of the Community's railway lays down that management of the railway infrastructure for mainline passenger operations shall be separated from train operation. This creates an interface between these two entities.

This annex specifies the requirements for management of EMC at this interface.

B.2 Definitions

For the purpose of this annex only the following definitions apply.

B.2.1
infrastructure controller
person or body that holds legal responsibility in respect of the railway network infrastructure (see B.5.1)

B.2.2
traction and rolling stock (T&RS)
moving vehicles of the railway and includes wagons, locomotives, coaches and multiple units. Vehicles used for service and maintenance duties are included

B.2.3
train operator
body which operates traction and rolling stock

B.2.4
failure
inability of an item of equipment to continue to perform its intended function

B.3 Assessment of risks of failures

The assessment and management of risk of failures arising out of the EM interaction between the electrical systems of trains and of infrastructure under normal operating conditions (see B.5.3) shall include the following steps.

- Identification of potential failures arising out of the EM interaction between infrastructure and T&RS.
- Identification of infrastructure EM immunity and emission levels relevant to the potential failures.
- Identification of the T&RS EM immunity and emission levels relevant to the potential failures.
- Analysis of the potential failures in sufficient depth to expose the key issues.
- Specification of control measures required to maintain the risks at acceptable levels.
- Demonstration of acceptability of the resulting failure levels.

© BSI 12-2000

B.4 Requirements

B.4.1 Reliability

The infrastructure controller and the train operators shall take all reasonable measures to ensure that under normal operating conditions their actions and equipment do not compromise reliability due to EMI.

B.4.2 Immunity

The train operator and the infrastructure controller shall take all reasonable measures to ensure that their respective systems, the trains, the power supply system, the signalling system and the telecommunications system are immune to reasonable levels of EM emission and those produced by portable transmitters whose use is permitted within the railway boundary.

B.4.3 Emission

B.4.3.1 EM emission from trains can cause failures of the infrastructure. EM emission from the infrastructure can cause failures of trains.

B.4.3.2 The infrastructure controller, in the course of an assessment of risks of failures, may require to obtain from the train operator details of EM emission of trains. In a like manner the train operator, in the course of an assessment of risks of failures, may also require to obtain from the infrastructure controller details of EM emission from the infrastructure. The train operator and the infrastructure controller shall, when reasonably required so to do, co-operate with each other to obtain test results or credible predictions of the relevant parameter values of voltage, current, magnetic field and electric field which are produced by the train operator's trains and/or the infrastructure controller's infrastructure when energized, at positions which are identified by the infrastructure controller and/or the train operator.

B.4.3.3 If appropriate locations for values of magnetic and electric fields have not been identified, values may be provided at positions appropriate to the hazard.

B.4.3.4 The field strengths for emission of radio noise within the railway boundary shall be agreed between the infrastructure controller and the train operator.

B.4.3.5 The limits for the induction of psophometric voltage into telecommunication links within the railway shall be agreed between the train operator and infrastructure controller, taking note of the International Telecommunications Union (ITU-T) limits.

B.5 Infrastructure and normal operating conditions

B.5.1 Infrastructure

The fixed equipment containing, among others, the following systems:

- power supply system, comprising power supply, cable, including connection to the supply industry and the conductors along the railway track, etc.;
- signalling system, comprising signalling, including track circuits, cables, interlockings and control rooms, etc.;
- telecommunication system, comprising telecommunications, including cables, phones, radio links and data communications, etc.;
- permanent way system.

© BSI 12-2000

B.5.2 Railway electrical system

The railway electrical system which comes under the EMC scope of this standard is highly complex and is made up of:

- the power supply system;
- signalling and telecommunication systems;

connected by:

- the running rails;
- the trains.

B.5.3 Normal operating conditions

These include the following:

- nominal conditions;
- operation under emergency traction conditions where currents may be greater than usual;
- the passage of trains across gaps in the power supply system such as at neutral sections and conductor rail gaps;
- raising and lowering pantographs;
- transient behaviour of T&RS such as inrush current into a.c. transformers and into d.c. filters;
- the production of transients and surges due to switching actions on the power supply system, on-train loads and/or the feed from the national grid;
- the electrical resonant behaviour of a.c. power supply systems;
- such partial failure or loss of connections between parts of a system as shall be agreed with the train operator;
- the effects of the range of weather conditions appropriate for the location;
- other conditions which may be expected due to the normal operation of the railway.

B.5.4 Assumptions

When assumptions and approximations are made, these shall be declared.

© BSI 12-2000

BSI — British Standards Institution

BSI is the independent national body responsible for preparing British Standards. It presents the UK view on standards in Europe and at the international level. It is incorporated by Royal Charter.

Revisions

British Standards are updated by amendment or revision. Users of British Standards should make sure that they possess the latest amendments or editions.

It is the constant aim of BSI to improve the quality of our products and services. We would be grateful if anyone finding an inaccuracy or ambiguity while using this British Standard would inform the Secretary of the technical committee responsible, the identity of which can be found on the inside front cover. Tel: 020 8996 9000. Fax: 020 8996 7400.

BSI offers members an individual updating service called PLUS which ensures that subscribers automatically receive the latest editions of standards.

Buying standards

Orders for all BSI, international and foreign standards publications should be addressed to Customer Services. Tel: 020 8996 9001. Fax: 020 8996 7001.

In response to orders for international standards, it is BSI policy to supply the BSI implementation of those that have been published as British Standards, unless otherwise requested.

Information on standards

BSI provides a wide range of information on national, European and international standards through its Library and its Technical Help to Exporters Service. Various BSI electronic information services are also available which give details on all its products and services. Contact the Information Centre. Tel: 020 8996 7111. Fax: 020 8996 7048.

Subscribing members of BSI are kept up to date with standards developments and receive substantial discounts on the purchase price of standards. For details of these and other benefits contact Membership Administration. Tel: 020 8996 7002. Fax: 020 8996 7001.

Copyright

Copyright subsists in all BSI publications. BSI also holds the copyright, in the UK, of the publications of the international standardization bodies. Except as permitted under the Copyright, Designs and Patents Act 1988 no extract may be reproduced, stored in a retrieval system or transmitted in any form or by any means – electronic, photocopying, recording or otherwise – without prior written permission from BSI.

This does not preclude the free use, in the course of implementing the standard, of necessary details such as symbols, and size, type or grade designations. If these details are to be used for any other purpose than implementation then the prior written permission of BSI must be obtained.

If permission is granted, the terms may include royalty payments or a licensing agreement. Details and advice can be obtained from the Copyright Manager. Tel: 020 8996 7070.

BSI
389 Chiswick High Road
London
W4 4AL